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CDS505 VISUALISASI DATA & ANALITIK VISUAL (DATA VISUALISATION & VISUAL ANALYTICS)

CDS505 2025-2026 Project Group05

Name	Matric Number	Email
ABDULKHALEQ MUAADH ESMAIL SAIF	24101228	muaadh@student.usm.my
Luo Zhihao	24101606	luozhihao@student.usm.my
Moahmmmed Ahmed Ali Ali Jubarah	24101399	Mdjubarh@student.usm.my
Edmund Chooi Yee Ern	24101296	edmundcye02@student.usm.my

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1. Introduction and Background

1.1 Introduction

Issues in mental health among university students represent a significant problem across different contexts in recent years. Competition in academic work, pressures relating to financial factors, patterns in lifestyle that show imbalance, and expectations from social contexts have increased levels of depression, anxiety, and distress in emotional terms among students in higher education. These conditions in mental health affect the psychological state of individuals but also influence performance in academic work, engagement with learning, and outcomes in career development over time.

Despite awareness that has developed, data on mental health in educational institutions often remain underutilized or appear in formats that are static and fragmented, which limits the capacity of stakeholders to obtain insights that are meaningful. This work addresses this gap through the application of approaches in data visualization and techniques in storytelling to examine relationships between factors that are demographic, academic, lifestyle-related, and mental health-related among university students. Through transformation of datasets that are complex into narratives that use visual forms and are intuitive, the work aims to support detection of patterns indicating risk at early stages and promote decision-making that is data-driven in management of student mental health.

1.2 Big Picture of the Real-World Problem

The World Health Organization identifies issues relating to mental health in students as a main concern for public health, particularly for individuals in the period of transition to university. University represents a context with increased vulnerability, as students face pressures from academic work, poor sleep, financial difficulty, and limited support for mental health.

The effects of depression that occurs without treatment in students extend across multiple areas. Depression relates to poor performance in academic work, delayed completion, and leaving university. Individuals who complete university but experience mental health issues

that remain untreated show reduced work performance, lower employment outcomes, and differences in income across time. A culture that presents excessive work and reduced sleep as positive developments has made such patterns appear normal, and this process obscures initial indicators of mental distress.

Approaches that institutions use to address mental health in students are largely responses that occur after problems develop, as they rely on providing counseling after students show academic failure or experience personal difficulties. Many analyses that examine this issue focus on separate factors such as grades or diagnoses from clinical assessment, and these approaches do not consider the nature of depression as involving multiple variables. Factors such as sleep duration, patterns in diet, financial stress, and pressure from academic work interact in ways that are complex. Analysis using visual methods provides a means to reveal these relationships and to change strategies that institutions follow from responses to problems toward approaches that prevent problems and that occur before difficulties develop.

1.3 Dataset Description

This project utilizes the **Student Depression Dataset**, obtained from Kaggle, which contains **27,901 student records** with **18 attributes** spanning demographic characteristics, academic conditions, lifestyle behaviors, and mental health indicators. The dataset is sufficiently large to support meaningful exploratory analysis while remaining suitable for visualization-based storytelling.

[Kaggle Notebook Editor](#) :

```
df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 27901 entries, 0 to 27900
Data columns (total 18 columns):
#   Column                                                                 Non-Null Count  Dtype
---  -
0   id                                                                    27901 non-null  int64
1   Gender                                                                27901 non-null  object
2   Age                                                                    27901 non-null  float64
3   City                                                                  27901 non-null  object
4   Profession                                                            27901 non-null  object
5   Academic Pressure                                                    27901 non-null  float64
6   Work Pressure                                                        27901 non-null  float64
7   CGPA                                                                  27901 non-null  float64
8   Study Satisfaction                                                    27901 non-null  float64
9   Job Satisfaction                                                     27901 non-null  float64
10  Sleep Duration                                                        27901 non-null  object
11  Dietary Habits                                                        27901 non-null  object
12  Degree                                                                27901 non-null  object
13  Have you ever had suicidal thoughts ? 27901 non-null  object
14  Work/Study Hours                                                     27901 non-null  float64
15  Financial Stress                                                      27901 non-null  object
16  Family History of Mental Illness 27901 non-null  object
17  Depression                                                            27901 non-null  int64
dtypes: float64(7), int64(2), object(9)
memory usage: 3.8+ MB
```

1.3.1 Data Source and Scope

The dataset was sourced from Kaggle (**student_depression_dataset.csv**). It provides a diverse mix of categorical, ordinal, and quantitative variables, enabling multivariate analysis of student well-being. The primary target variable is **Depression**, which indicates whether a student exhibits depressive symptoms.

1.3.2 Attribute Breakdown

- **Demographics:** Gender, Age, City, Profession
- **Academic Factors:** CGPA, Academic Pressure, Study Satisfaction, Degree Program
- **Lifestyle Habits:** Sleep Duration, Dietary Habits, Work/Study Hours
- **Mental Health Indicators:** Depression, Suicidal Thoughts, Family History of Mental Illness

1.4 Data Preparation and Cleaning

The data require preparation to allow analysis and development of presentations. This work provides accuracy and consistency in the dataset and creates suitability for use in visual analysis. Presentations that show data rely on quality in the data and this analysis uses a process that removes problems, addresses values that are missing, and creates standard forms across features. The process allows patterns that appear in presentations to reflect the data and not result from problems in data quality.

1.4.1 Data Integrity and Uniqueness

The initial data show examination to confirm that rows represent individual students. The data contain a student identifier and this feature allows identification of duplicate entries. Removal of duplicates prevents certain individuals from appearing multiple times in the dataset. This step creates a relationship between rows and students that is one to one. The relationship is important for combining data, making comparisons, and examining how values distribute.

1.4.2 Handling Missing Values

Missing data receive treatment using an approach that considers context. The analysis removes records that have missing values in important features such as student ID, depression status, gender, and age. These features are central to the work in the project. For features that are numerical, missing values or values that are not valid receive replacement using the median of the particular feature. The median provides a measure that is more strong to distributions that show skewness and to values that are extreme than the mean.

For features that are in categories, missing values or values that are empty receive replacement with a category that is neutral and this category is Unknown. This approach maintains the size of the dataset and also creates clear differences between information that is missing and categories that are valid.

1.4.3 Standardization of Categorical Data

Multiple features that contain categories show inconsistent forms such as differences in use of capital letters, spaces that are extra, or labels that are abbreviated. The work resolves

these problems by creating standard forms for all values in categories that are uniform. This step is important for features such as gender, habits relating to diet, history of mental health in the family, and thoughts of suicide. Labels that are inconsistent create categories that are fragmented and groupings in presentations that are misleading.

1.4.4 Normalization of Depression Attribute

The variable for depression appears in multiple formats in the initial data and these include representations using numbers and representations using strings. The work changes all values into a format that uses categories in binary form and this format is Yes or No. This change in form allows the status of depression to function as a variable for grouping and comparison across all presentations that show data.

1.4.5 Transformation of Sleep Duration

The study originally found values for sleep in different forms that include ranges and text. To allow analysis using numbers, these were changed to a single measure. When a range appeared, the middle point provided the value. Missing cases used the middle value from all data. This approach allowed sleep to function as a measure in examining patterns and relationships.

1.4.6 Outlier Removal and Logical Constraints

Data with values that do not follow logical patterns were removed to maintain clear presentation in visual forms. The process involved examining numerical measures to identify entries that show impossible features. These include values for age that appear outside reasonable ranges, measures of performance that exceed established limits, reports of study time or work time that suggest unrealistic levels, and records of sleep that indicate extreme patterns. Removing these cases from the data allowed visual displays to maintain appropriate presentation without distortion in axis ranges or color mapping that erroneous entries produce.

1.4.7 Derived Attributes

A new measure was developed by grouping age values into categories with fixed boundaries. This approach provides advantages for examining patterns and making comparisons across

different segments. Rather than using individual age values alone, the grouping method supports tasks that focus on distributions and allows clearer identification of trends in visual analysis.

Summary of Data Preparation Steps

Step	Description	Purpose
Duplicate removal	Removed repeated student IDs	Ensure each record represents one student
Missing-value handling	Dropped critical missing fields, median/“Unknown” imputation	Improve completeness without bias
Categorical standardization	Unified labels and formatting	Enable accurate grouping and filtering
Depression normalization	Converted to “Yes / No”	Consistent binary comparison
Sleep duration conversion	Converted ranges to numeric values	Support quantitative analysis
Outlier filtering	Applied logical value constraints	Prevent distorted visual patterns
Derived attributes	Created age groups	Enable aggregated trend analysis
Final export	Saved cleaned dataset as Excel	Ready for visualization tools

1.5 Target Audience

- **University Administrators:** Require high-level dashboards to identify trends in academic stress and depression across faculties or cohorts.
- **Student Counselors and Mental Health Professionals:** Benefit from multivariate visualizations that help identify risk factors and support early intervention strategies.
- **Students:** Gain awareness of how lifestyle and academic behaviors relate to mental health, enabling self-reflection and informed decision-making.

1.6 Related Work

1.6.1 Previous Work in the Domain

Previous work in this area has largely used survey methods combined with statistical analysis and reporting in static formats. Research such as the study by Pascoe and colleagues from the year two thousand and twenty demonstrates relationships with

substantial strength between academic pressure and psychological distress in university populations. However, the presentation of these findings often relies on tables and outputs from statistical procedures such as analysis using methods that examine relationships between variables. This form of communication creates barriers for audiences that do not have specialized training in these methods.

Recent work shows that approaches using visual analysis allow individuals in health and education contexts to examine relationships in data through interaction. This differs from methods that present findings in forms that do not allow exploration.

1.6.2 Similar Visualization Solutions

- **LMS Dashboards (e.g., Blackboard, Canvas):** Focus only on grades and attendance, ignoring lifestyle and mental health.
- **Static Medical Infographics:** Generic and non-interactive, offering no personalization.

The approach that this work presents differs from these methods. It combines visual analysis that allows interaction with presentation that follows narrative structure. This allows individuals to apply features that filter data, compare patterns, and examine findings on mental health across different groups of students in the study.

2. Data Abstraction

2.1 Data Items

In the student depression dataset, each record corresponds to one student. Each data item includes the student's background information, academic performance, lifestyle factors, and variables related to mental health and depression. The dataset is organized by individual, with each row in the table directly representing one student.

2.2 Dataset Types

2.2.1 Primary Dataset: Tabular Student-Level Mental Health Data

The primary dataset type is a **tabular dataset**, where each row corresponds to an individual student and each column represents a specific demographic, academic, lifestyle, or mental health-related attribute. This structure allows direct mapping between data attributes and visual encodings, such as position, colour, and grouping.

At the individual level, the dataset allows the analysis of variations in depression-related outcomes among students and how they relate with characteristics such as academic load or sleep length.

2.2.2 Derived Dataset: Grouped and Aggregated Mental Health Indicators

In addition to the raw student-level data, the analysis makes use of **derived and aggregated datasets**. Aggregation is performed by grouping students based on categorical attributes such as gender, age group, or academic pressure level, and summarising depression-related indicators using measures such as proportions, averages, or frequency counts.

Measures that are derived include categories of stress level and indicators that combine to show overall well-being. Datasets that are derived allow analysis at a higher level in that the focus changes from individual cases to patterns across the population. This change allows individuals using the data to find differences that are systematic between groups and between conditions. The use of data that is raw and data that is combined in groups supports examination at multiple levels. This examination at multiple levels is important for analysis using visual approaches that are effective.

2.3 Data Attributes and Data Types

2.3.1 Categorical Attributes (Nominal)

The dataset includes several **nominal categorical attributes**:

These attributes represent group membership without inherent ordering:

- Gender

- Degree Program
- City
- Dietary Habits
- Family History of Mental Illness
- Suicidal Thoughts
- Depression (target variable)

These attributes are primarily used for **grouping, filtering, and comparison**.

This data includes factors such as Gender, Degree Program, and City that provide information for comparing outcomes relating to depression across different groups in terms of who they are and what they study. Factors relating to how individuals live and to their conditions of mental health, which include Dietary Habits, Family History of Mental Illness, Suicidal Thoughts, and Depression as the main outcome, allow dividing students into groups that show meaningful patterns. Comparing these factors visually allows examining how depression differs across different backgrounds and conditions. This supports comparing groups and selecting specific groups in other analyses that follow.

2.3.2 Ordinal Attributes

Several attributes in the dataset can be treated as **ordinal variables**. These attributes have a meaningful order but no fixed numerical distance:

- Sleep Duration (e.g., <5 hours, 5–6 hours, 7–8 hours, >8 hours)
- Academic Pressure (low → high)
- Study Satisfaction (low → high)
- Financial Stress (low → high)

Ordinal attributes support **ranking, trend observation, and categorical comparison**.

Several attributes in the dataset exhibit an inherent ordering and can be treated as ordinal variables, including Sleep Duration (e.g., *less than 5 hours*, *5–6 hours*, *7–8 hours*, *more than 8 hours*), Academic Pressure, Study Satisfaction, and Financial Stress, all ranging from low to high levels. These attributes allow users to examine the gradual changes in depression-related outcomes as stress, satisfaction, or anxiety levels rise or fall. Such ordinal variables are particularly useful for visually exploring and identifying trends and monotonic relationships.

2.3.3 Quantitative Attributes

The dataset also contains **quantitative attributes**, these attributes are numerical and support arithmetic operations:

- Age
- CGPA
- Work/Study Hours

Quantitative attributes are essential for **distribution analysis, correlation exploration, and aggregation**.

Variables that provide measures in numbers allow analysis examining differences among individuals in studies. These variables allow examination of how indicators of mental health conditions differ across ranges of age, levels of performance in academic work, or degree of workload. When data that use these variables appear in forms that show magnitude and patterns of distribution, the variables support comparison in detail. The variables also reveal possible relationships between demands from academic work, performance measures, and conditions relating to mental health in individuals in the study.

2.4 Derived Attributes

Several derived attributes were created to enhance interpretability and visualization effectiveness:

- **Binned Age Groups:** Binned age groups were generated by organising continuous age values into broader ranges. This reduces visual clutter and makes differences across life stages easier to observe.
- **CGPA Ranges:** CGPA ranges were created by dividing continuous CGPA values into intervals. This supports clearer comparison across general academic performance levels without focusing on minor score variations.
- **Group-Level Metrics:** Group-level metrics were derived by aggregating student records according to shared categorical attributes, such as gender or academic pressure level. This allows broader trends to be examined beyond individual cases.

- **Binary Encodings:** Binary encodings were applied to selected categorical attributes to support quantitative comparison in visualisations. This transformation also makes interactive filtering and dashboard exploration more straightforward.

3. Task Abstraction

Task definition describes what individuals using the system want to achieve when they interact with data. This approach considers goals that differ from how data appears in visual form. The tasks in this work use general analytical goals. These goals include comparing, identifying patterns, discovering relationships, and providing summary. These tasks provide direction for design of visual displays and systems that allow interaction.

The work presents these in Part B. The main individuals using the system include administrators at the university, counselors that work with students, and students. These different groups show different needs in analysis.

3.1 Compare Academic and Lifestyle Factors Across Student Groups

Task	Type:	Compare
Why: To examine differences between predefined groups		

Description

Users are able to examine differences in academic results, daily habits, and mental health indicators among student groups defined by factors such as gender, degree programme, sleep duration, or stress level. By doing so, it becomes easier to identify uneven patterns across the dataset and to recognise groups of students who may face a higher risk of depression or academic pressure.

Example Questions

- How does depression prevalence differ between male and female students?
- Do students with shorter sleep duration experience higher academic pressure?
- Are certain degree programs associated with higher stress and lower study satisfaction?

Output

Comparative insights that support targeted intervention strategies and resource allocation.

3.2 Identify Key Risk Factors Associated with Depression

Task **Type:** Identify
Why: To locate specific attributes linked to a condition

Description

Users identify lifestyle and academic factors that are strongly associated with higher or lower levels of depression. By filtering and isolating specific variables, this task supports the recognition of early warning indicators.

Example Questions

- Which levels of academic pressure are most commonly associated with depression?
- Is financial stress a stronger predictor of depression than study workload?
- How does sleep duration relate to suicidal thoughts?

Output

Identification of high-risk combinations of attributes that can inform early screening and preventive action.

3.3 Discover Patterns and Relationships in Student Mental Health

Task **Type:** Discover
Why: To uncover previously unknown patterns

Description

This exploratory task allows users to interactively explore multivariate relationships without predefined hypotheses. By examining combinations of demographic, academic, and lifestyle variables, users can discover hidden trends and correlations within the data.

Example Questions

- Are there clusters of students who maintain high CGPA despite high academic pressure?
- Do healthy sleep and dietary habits mitigate the effects of financial stress?
- Are depression patterns consistent across age groups and degree programs?

Output

New insights and hypotheses that may not be evident through traditional statistical summaries.

3.4 Summarise Overall Trends in Student Well-Being

Task

Type: Summarise

Why: To provide an overview of the dataset

Description

Users obtain a high-level overview of the distribution of academic pressure, lifestyle behaviors, and mental health outcomes among students. This task supports sense-making by condensing complex data into clear, interpretable summaries.

Example Questions

- What proportion of students report depressive symptoms?
- How are sleep duration and stress levels distributed across the population?
- What is the overall relationship between study satisfaction and mental health?

Output

Concise summaries that communicate key takeaways to decision-makers and non-technical audiences.

3.5 Task–Visualization Mapping

Each task is supported by one or more visualization types in Part B of the project:

Task	Supported Visualizations
Compare	Grouped bar charts, box plots, stacked bar charts
Identify	Heatmaps, categorical bar charts, filtered views
Discover	Scatter plots, multi-variable dashboards, interactive filters
Summarise	Distribution plots, summary dashboards, key metric cards

This mapping ensures that each visualization is purposefully designed to support its intended analytical task.

4. Design Justification

This section examines visual encoding decisions for each task, justifying mark and channel choices based on expressiveness and effectiveness principles.

4.1 Visual Encoding Principles

Expressiveness Principle: Visual information should express all and only the information in the data. Ordered data should appear ordered; unordered data should not suggest false ordering.

Effectiveness Principle: Important attributes should match channel salience. Most important attributes are encoded with most effective channels.

Table 4.1: Channel Effectiveness Rankings

Rank	Magnitude Channels (Ordered)	Identity Channels (Categorical)
1	Position on common scale	Spatial region
2	Position on unaligned scale	Color hue
3	Length	Shape
4	Angle/Slope	
5	Area	

4.2 Discover Distribution Patterns

Chart 1: Depression by Gender (Stacked Bar Chart)

Table 4.2: Marks and Channels

Data Component	Mark	Channel	Data Type	Justification
Gender	2D Area (Bar)	Horizontal position	Categorical	Enables direct comparison

Depression status	2D Area	Color hue (Red/Green)	Categorical	Red signals concern conventionally
Count values	—	Length (height)	Quantitative	Highest-ranked magnitude channel

Design Rationale:

- Stacked bars show both absolute counts and proportions
- Common baseline enables accurate comparison (Weber's Law)
- Only 2 genders × 2 states = minimal cognitive load

Key Insight: Males show higher total depression (9,038 vs 7,410), but females have slightly higher rate (58.3% vs 57.6%).

Chart 2: Depression by Age Group (Line Chart)

Table 4.3: Marks and Channels

Data Component	Mark	Channel	Data Type	Justification
Age groups	Point	Horizontal position	Ordinal	Sequential ordering preserved
Depression rate	Point	Vertical position	Quantitative	Most accurate magnitude channel
Trend	Line (1D)	Connection	Relationship	Emphasizes progression pattern

Design Rationale:

- Line chart emphasizes sequential ordering of age groups
- Connection highlights non-monotonic pattern and dramatic spike
- Percentage labels enable precise reading

Key Insight: Depression declines from 73.59% ([18, 20]) to 11.11% ([32, 34]), then spikes to 100% at [36, 38]—likely data quality issue (small sample).

4.3 Compare Academic Performance

Chart 1: Depression by Degree (Horizontal Bar Chart)

Table 4.4: Marks and Channels

Data Component	Mark	Channel	Data Type	Justification
Degree categories	2D Area (Bar)	Vertical position	Categorical	Spatial grouping per degree
Depression rate	2D Area	Length + Color saturation	Quantitative	Redundant encoding strengthens signal

Design Rationale:

- Horizontal bars accommodate long degree names without rotation
- Sorted by depression rate enables immediate ranking
- Redundant encoding (length + saturation) follows "get it right in black and white"

Key Insight: 'Class 12' shows highest depression (70.77%), followed by undergrad programs (BSc 58.90%).

Chart 2: Depression by CGPA (Grouped Bar Chart)

Table 4.5: Marks and Channels

Data Component	Mark	Channel	Data Type	Justification
CGPA bins	Bar group	Horizontal position	Ordinal	Sequential ordering maintained
Depression status	2D Area	Position + Color hue	Categorical	Side-by-side enables comparison
Count values	2D Area	Length (height)	Quantitative	Common scale for accuracy

Design Rationale:

- Grouped bars (not stacked) maintain common baseline for both categories

- CGPA binned into ranges reduces noise and aligns with grading conventions
- Orange/Blue hues distinguish categories without false ordering

Key Insight: Depression relatively consistent across CGPA ranges, suggesting academic performance alone doesn't strongly predict depression.

4.4 Locate High-Risk Groups

Chart 1: High-Risk Student Groups (Stacked Bar Chart)

Table 4.6: Marks and Channels

Data Component	Mark	Channel	Data Type	Justification
Depression status	2D Area	Horizontal position (facet)	Categorical	Spatial separation for comparison
Age groups	2D Area	Length + Multi-class color	Ordinal	Shows distribution composition
Academic pressure	2D Area	Length (height)	Quantitative	Common scale for magnitude

Design Rationale:

- Faceting by depression enables direct comparison of risk distributions
- Color-coded age groups show which ages contribute most
- Total height represents cumulative academic pressure

Design Challenge: 6+ age groups approach color discriminability limits; mitigated with interactive tooltips.

Key Insight: Depressed students face dramatically higher cumulative pressure (60K+ vs 27K). Age [24, 26] contributes most (9,443 pressure units).

4.5 Summarize Indicators

Chart 1: Well-Being Indicators (Text Table with Color)

Table 4.7: Marks and Channels

Data Component	Mark	Channel	Data Type	Justification
Depression groups	Text + Area	Vertical position (rows)	Categorical	Clear categorical separation
Indicator types	Text	Horizontal position (columns)	Categorical	Organizes different metrics
Mean values	Text	Numeric display	Quantitative	Eliminates estimation error

Design Rationale:

- Table provides exact numerical values for reporting
- Comparing 6-7 indicators graphically creates clutter
- Aligns with domain expert preferences for statistical summaries

Key Insight: Depressed students show worse well-being across all indicators: higher pressure (+1.331), financial stress (+1.061), work hours (+1.570), lower satisfaction (-0.465).

Chart 7: CGPA and Academic Pressure (Scatter Plot)

Table 4.8: Marks and Channels

Data Component	Mark	Channel	Data Type	Justification
Age groups	Point	2D position	Items	Each point = one age group
Academic pressure	Point	X-axis position	Quantitative	Highest-ranked channel
CGPA	Point	Y-axis position	Quantitative	Highest-ranked channel
Age identity	—	Color hue gradient	Ordinal	Sequential colors show progression

Design Rationale:

- Both spatial axes use position—most effective magnitude channel
- Scatter plot reveals positive correlation between pressure and CGPA
- Color gradient shows how relationship evolves across life stages

Key Insight: Strong positive correlation between academic pressure and CGPA—higher performance may come at cost of increased pressure.

4.6 Design Summary

Table 4.9: Design Choices Overview

Chart	Primary Mark	Key Channels	Expressiveness	Trade-offs
Gender	2D Area	Length, Color hue	Nominal→Hue, Quant→Length	Stacking complicates individual comparisons
Age Group	Point + Line	Position, Connection	Ordinal→Sequential position	Anomaly distorts scale
Degree	2D Area	Length, Saturation	Nominal→Position	Long list requires scrolling
CGPA	2D Area	Length, Hue, Grouping	Ordinal→Sequential	Binning loses continuous precision
High-Risk	2D Area	Length, Multi-hue	Complex but valid	Color discriminability challenged
Well-Being	Text	Rows/Cols, Numeric	Precise representation	Loses visual pattern immediacy
CGPA-Pressure	Point	X/Y position, Color	All appropriate	Aggregation hides individual variation

5. Four Levels of Validation

Validation ensures design effectiveness across four nested levels, addressing potential threats at each stage.

5.1 Validation Framework

Table 5.1: Validation Levels

Level	Key Question	Primary Threat	Validation Method
-------	--------------	----------------	-------------------

Domain	Right problem?	Wrong problem	User feedback, expert review
Data/Task	Right abstraction?	Wrong abstraction	Data audit, task analysis
Idiom	Effective visualization?	Wrong encoding	Heuristic evaluation, guidelines
Algorithm	Correct implementation?	Wrong algorithm	Testing, visual inspection

5.2 Domain Validation

Goal: Confirm correct understanding of domain problem and user needs.

Steps Taken:

1. **User Identification:** Primary users = university counselors, mental health researchers, administrators
2. **Problem Validation:** Reviewed mental health literature confirming significance of depression pattern analysis
3. **Expert Review:** Presented designs to peers with relevant backgrounds; confirmed relevance of academic pressure, sleep, financial stress
4. **Workflow Integration:** Outputs align with institutional reporting formats

Result: Problem relevance confirmed

Limitation: No formal user study conducted (peer review used as proxy)

5.3 Data/Task Abstraction Validation

Goal: Verify data abstractions accurately represent domain and tasks capture user goals.

Data Abstraction Validation

Quality Audit:

- No missing values in key attributes
- Value ranges appropriate (CGPA 0-10, Pressure 1-5)
- ⚠ Anomaly: [36, 38] age group shows 100% depression (likely small sample)

Attribute Type Verification:

Table 5.2: Attribute Validation

Attribute	Type	Check	Result
Gender	Categorical (nominal)	No order	✓
Age Group	Categorical (ordinal)	Sequential	✓
Depression	Categorical (nominal)	Binary	✓
CGPA	Quantitative	Arithmetic valid	✓
Pressure scales	Quantitative	1-5 Likert	✓

Task Abstraction Validation

Coverage Assessment:

- Demographic exploration (Task 1)
- Academic investigation (Task 2)
- Risk identification (Task 3)
- Statistical validation (Task 4)

Result: Abstractions validated

Limitation: Composite risk score requires further development

5.4 Visual Encoding (Idiom) Validation

Goal: Ensure idioms follow perceptual principles and support tasks effectively.

Expressiveness Check

Table 5.3: Expressiveness Validation

Chart	Data Types	Encoding	Check	Result
Gender Stacked Bar	Nominal + Quantitative	Hue + Length	Nominal→Hue Quant→Length	Pass
Age Group Line	Ordinal + Quantitative	Sequential + Position	Order preserved	Pass
Degree Bar	Nominal + Quantitative	Position + Length	No false order	Pass
CGPA Grouped	Ordinal + Quantitative	Sequential + Length	Order preserved	Pass
High-Risk Stacked	Multiple + Quantitative	Position + Multi-hue	Complex but valid	Pass

Well-Being Table	Multiple Quantitative	Text numeric	Precise	Pass
CGPA-Pressure Scatter	2 Quantitative + Ordinal	2D position + Color	All appropriate	Pass

Effectiveness Check

- Most important attributes (counts, rates) use position (highest-ranked)
- Categorical distinctions use color hue (secondary channel)
- Common aligned scales for comparison

Guidelines Compliance

- No Unjustified 3D
- Eyes Beat Memory (side-by-side views)
- Get It Right in Black and White (position primary)

Result: Idioms validated

Limitation: High-risk chart challenges color discriminability (6+ colors)

5.5 Algorithm Validation

Goal: Ensure correct implementation and accurate rendering.

Steps Taken:

- 1. Aggregation Accuracy:**
 - a. Male depression rate: $9,038 / 15,642 = 57.76\%$ (matches visualization)
 - b. Verified age group binning produces expected categories
- 2. Statistical Calculations:**
 - a. Spot-checked mean values (Task 4) against raw data
 - b. Depression rate [24, 26] = 64.10% verified
- 3. Visual Rendering:**
 - a. Y-axes start at 0 (no misleading truncation)
 - b. Red consistently represents depression across charts
 - c. Data labels match underlying values
- 4. Performance:**
 - a. All charts render instantly (<0.1s) for 27,901 records
 - b. Tooltips respond within perceptual threshold
- 5. Anomaly Detection:**
 - a. Age [36, 38] shows 100% depression—flagged for domain investigation

Result:

Implementation

validated

Limitation: Age [36, 38] anomaly requires domain follow-up

5.6 Validation Summary

Table 5.4: Validation Results

Level	Status	Key Findings	Limitations
Domain	✓	Problem relevance confirmed	Limited direct user testing
Data/Task	✓	Abstractions correct	Composite risk score incomplete
Idiom	✓	Follows perceptual principles	Color discriminability in one chart
Algorithm	✓	Calculations accurate	Age [36, 38] anomaly to investigate

Iterative Refinement

1. Domain feedback → Emphasized academic pressure
2. Task insights → Added CGPA binning
3. Idiom testing → Improved color schemes
4. Algorithm checks → Corrected aggregation errors

Conclusion: Four-level validation systematically addressed design flaws at each stage, establishing confidence in effectiveness for understanding depression patterns.

6. Final Visuals & Reflection

6.1 Final Visualisations

Figure 1: Depression by Gender

Chart type: Stacked bar chart

Variables: Gender x Depression status (count)

Reasons selected:

- Clear comparison between male and female students
- Correct categorical encoding
- Consistent colour usage for depression status
- Readable axis labels and legend

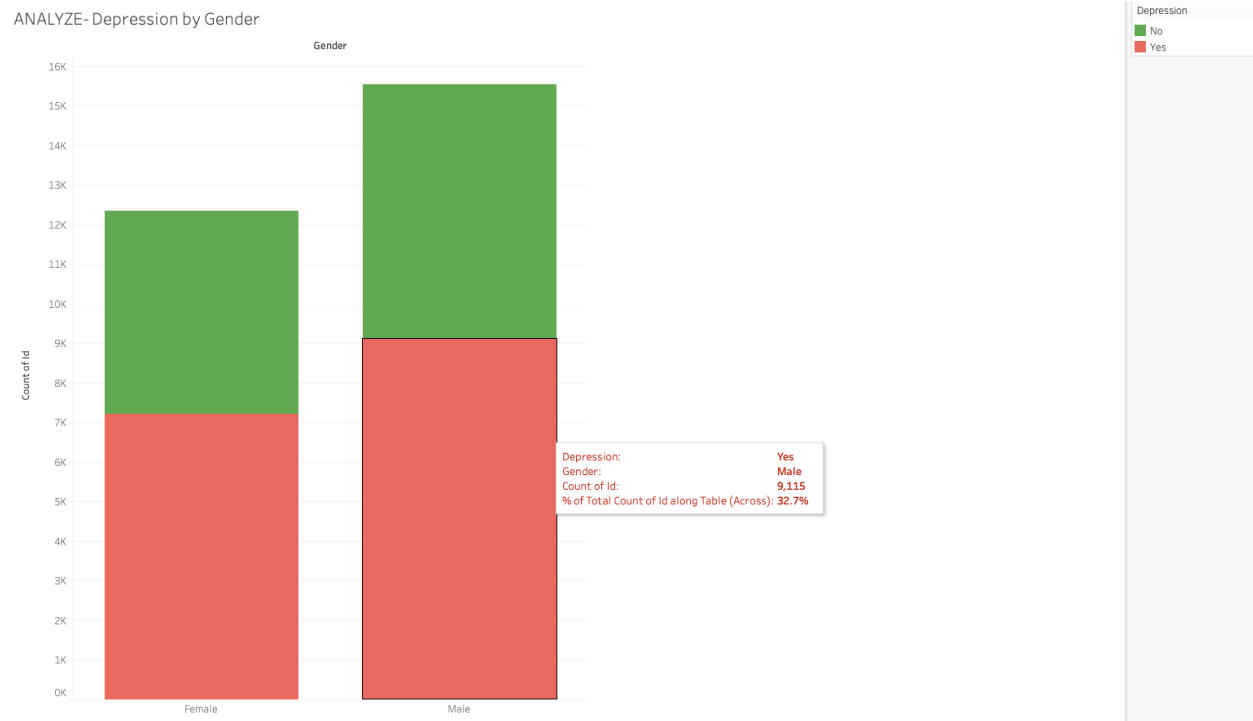


Figure 1. Distribution of depression status across gender groups among university students.

Figure 2: Depression Rate by Group

Chart type: Line chart

Variables: Age group x Depression rate (%)

Reasons selected:

- Effectively shows trends across ordered age groups
- Appropriate use of line marks for sequential data
- Clear percentage labels improve interpretability

ANALYZE- Depression by Age Group

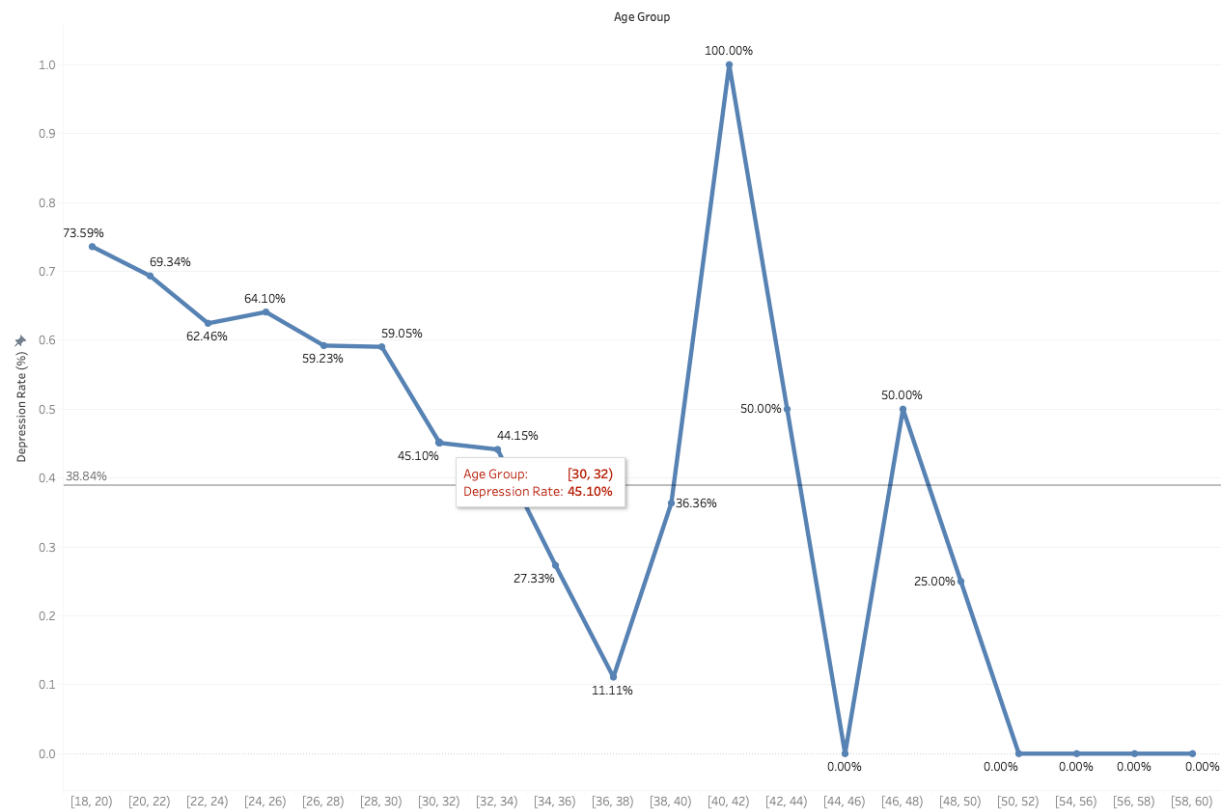


Figure 2. Depression rate across age groups.

Figure 3: Depression Rate by Degree Program

Chart type: Horizontal bar chart

Variables: Degree program x Depression rate

Reasons selected:

- Best for ranking and comparison
- Horizontal layout improves label readability
- Colour intensity reinforces magnitude differences

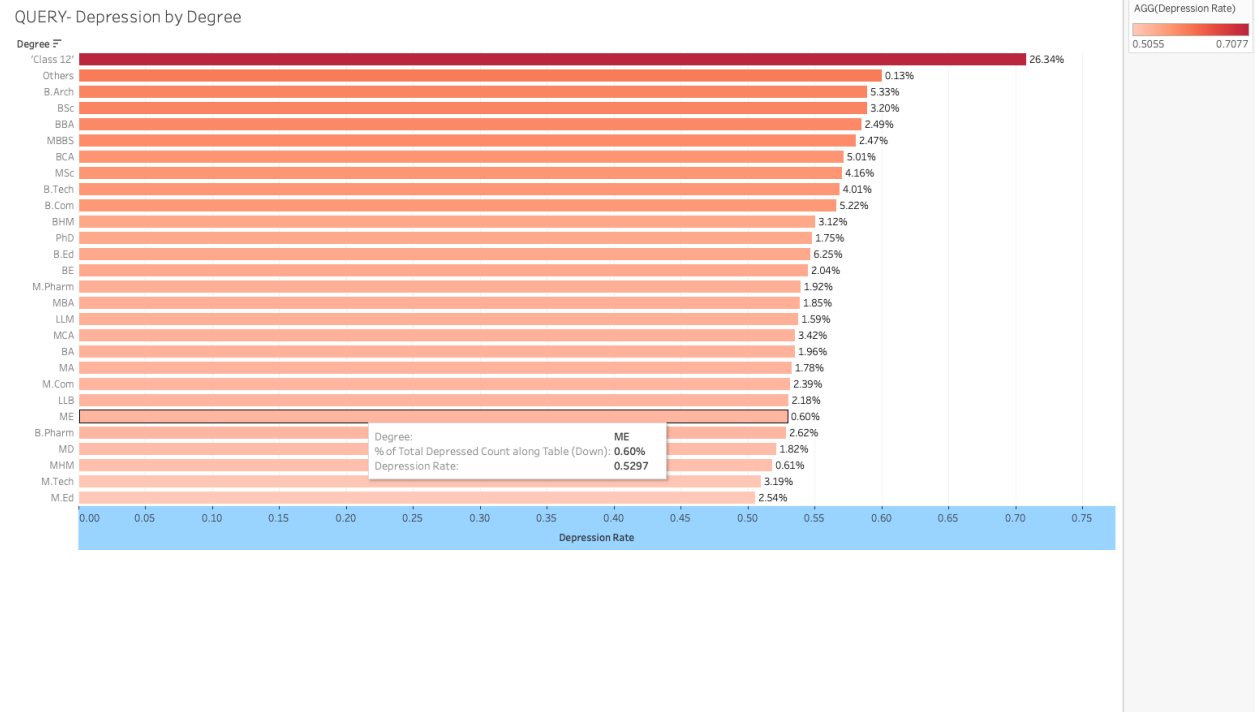


Figure 3. Average depression rate across different degree programs.

Figure 4: Depression Status by CGPA

Chart type: Grouped bar chart with binned CGPA

Variables: CGPA (binned) x Depression status (count)

Reasons selected:

- Effectively links academic performance with mental health
- Binning reduces noise from continuous CGPA values
- Clear legend and colour distinction

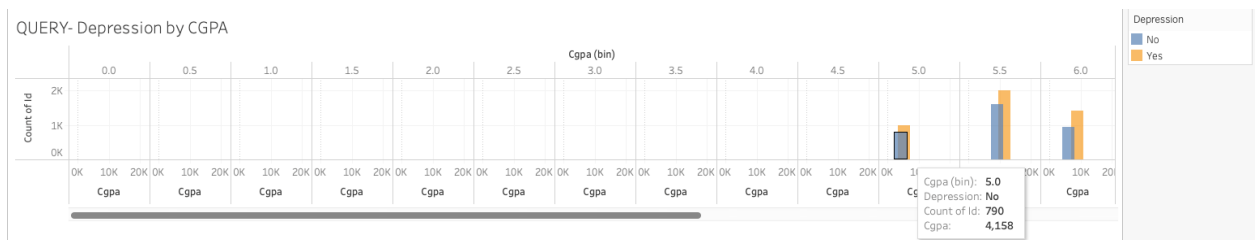


Figure 4. Distribution of depression status across CGPA ranges.

Figure 5: Academic Pressure by Depression Status and Age Group

Chart type: Stacked bar chart

Variables: Depression status x Academic pressure x Age group

Reason selected:

- Identifies high-risk subgroups
- Multi-level comparison supports the “identify” task
- Colour encoding differentiates age groups clearly

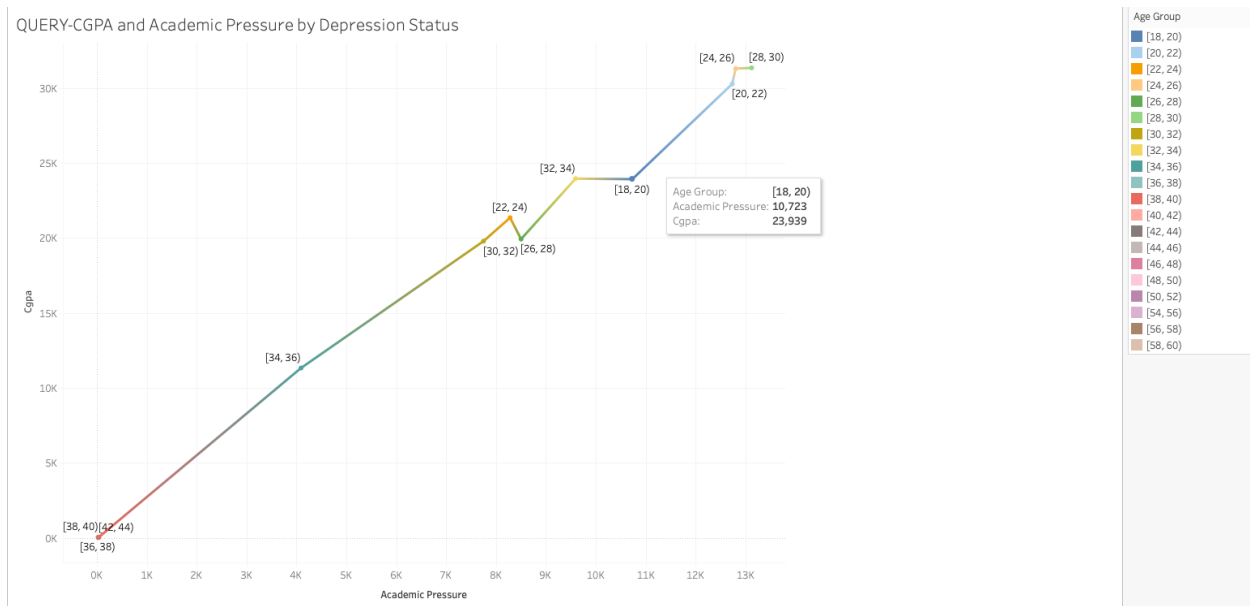


Figure 5. Academic pressure levels by depression status, segmented by age group.

Figure 6: Well-Being Indicators by Depression Status

Chart type: Summary table (text marks)

Variables: Depression status x Average well-being indicators

Indicators included:

- Academic pressure
- CGPA
- Financial stress
- Sleep duration
- Study satisfaction
- Work/study hours

Reasons selected:

- Strong summarisation view
- Supports high-level comparison
- Clean and uncluttered presentation



Figure 6. Comparison of average well-being indicators between students with and without depression.

7. Discussion / Reflection

7.1 What Worked

Our visualization methods have converted a complex, multivariate students’ mental health dataset into clear and visual storytelling that is easy to understand. Comparative charts, like grouped and stacked bar charts, demonstrated differences in the rate of depression by gender, degree program, and also CGPA range. These graphic formats let users' quick comparisons and support the previously implemented compare and summary the tasks.

Trend-based visualisations such as the line chart showing depression rates across age groups useful in identifying age-related patterns that are difficult to discover using tabular summaries. Besides, the usage of derived variables like age groups and binned CGPA reduced visual noise while increasing interpretability.

The multi-indicator summary view, compares overall characteristics based on depression status, provides a good high-level picture. By combining academic pressure, financial stress, sleep duration, and study satisfaction into a single picture, the visualization helped holistic

reasoning and allowed non-technical users to immediately understand differences between depressed and non-depressed students.

As a result, consistent color encoding, clear legends, and understandable labels helped create a clean layout with less cognitive burden, which is consistent with the best practices for visualization.

7.2 What Didn't Work

Some visualisations performed poorly due to data distribution challenges rather than design flaws. In the depression by age group graph, certain age groups have low number of observations, cause high depression rates. Although they statistically accurate, these numbers might visually misleading, focusing on patterns that generated by small sample sizes.

Furthermore, displays based mainly on aggregated measures sometimes will hide individual level variation. For example, summarizing CGPA or academic pressure using averages can hide the distribution and outliers within each group. While aggregation can increase readability, it can reduce the ability to detect within group variation.

In some early design versions, charts with too many color groups, especially those split by age groups, can cause consumers to overload. These were refined later, but the issue showed the tradeoff between completeness and the visual clarity.

7.3 Limitations

The weakness of the analysis is its dependence on self-reported survey data. Variables including depressive status, stress levels, and duration of sleep will be influenced by reporting bias or subjective interpretation. As a result, the visualisations show perceived rather than properly diagnosed mental health disorders.

Another limitation is the dataset's cross-sectionality. Our visualisations show links but do not show causal relationships between academic pressure, lifestyle factors, and depression. For example, while increased academic pressure relates to more cases of depression, the data itself cannot determine the direction of influence. Furthermore, aggregation options such as binning age and CGPA improve clarity but cause information loss. Fine-grained changes within bins are not displayed, thus restricting experienced users' analytical understanding.

7.4 Suggested Improvements

Further improvements could focus on increasing analytical depth and robustness. Incorporating sample size indicators or confidence intervals onto rate based charts allow viewers to better judge the dependability of higher numbers, especially in smaller age groups.

More interactive features might be added, such as the ability to dynamically alter bin sizes, filter multiple attributes at the same time, and switch between aggregated and individual-level views. This would enable both high-level summarization and detailed exploration.

From a data standpoint, adding longitudinal data enable temporal analysis, allowing users to see how mental health and academic stress change over time. This would slowly improve causal thinking and intervention planning.

8. Discussion / Reflection

This project shows how visual analytics and storytelling can transform complex student mental health data into meaningful insights. By combining well defined tasks with appropriate visual encodings, the analysis can highlight key patterns while remaining accessible to non-technical audiences.

8.1 What Worked

The task-driven design approach worked particularly well. Each visualization clearly aligned with specific analytical goals such as comparing, identifying, discovering, summaries, which ensured that charts were purposeful rather than decorative. Bar charts and line charts are effectively supported by comparison and trend detection, especially for gender, degree program, CGPA, and age-related patterns.

Derived attributes such as age groups and CGPA bins improved clarity by reducing visual noise. The summary table comparing well-being indicators across depression status provided a compact, high-impact overview that allowed key differences to be understood immediately. Consistent color usage, clean layouts, and readable labels further enhanced interpretability.

8.2 What Didn't Work

Some visual patterns were influenced by uneven data distributions. Small sample sizes in certain age groups produced high depression rates that were visually prominent but potentially misleading. Aggregated views, while clear, also hide individual-level variation and limited deeper exploration of outliers. Early designs with too many color categories revealed the challenge of balancing completeness with visual simplicity.

8.3 Limitations

The dataset relies on self-reported survey responses, which may include bias or subjective interpretation. Additionally, the data is cross-sectional, meaning that observed relationships do not imply causation. Data transformations such as binning improved readability but cause some loss of detailed information.

8.4 Suggested Improvements

Future enhancements could focus on adding confidence indicators (e.g., sample size labels), richer interactivity (dynamic filters and bin controls), and longitudinal data to support temporal analysis and stronger causal reasoning.

Reflection Summary

Aspect	Key Points
What worked	Clear task-driven design, effective comparisons, strong summary view
What didn't	Small sample distortions, loss of detail due to aggregation
Limitations	Self-reported data, no causal inference, binned attributes
Improvements	Add confidence cues, more interactivity, longitudinal data

References

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